

SAFETY DATA SHEET



ZINC TOP®

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	18.12.2025	50001138	Date of first issue: 18.12.2025

SECTION 1. IDENTIFICATION OF THE SUBSTANCE OR MIXTURE AND OF THE COMPANY OR UNDERTAKING

Chemical product identification : ZINC TOP®

Relevant identified uses of the substance or mixture and uses advised against

Recommended use : Crop nutrition

Restrictions on use : Use as recommended by the label.

Details of the supplier of the safety data sheet

Company name of supplier : FMC QUIMICA CHILE LTDA

Supplier's address : AVDA VITACURA 2670,
PISO 15, LAS CONDES,
VITACURA, SANTIAGO, CHILE
+56 2 28204200

E-mail address : SDS-Info@fmc.com

Emergency and toxicological information number in Chile : Chile: Spills: CITUC: +56 2 2247 3600 (24 hours) Fire: 132 (24 hours)
+56-22-5814934 (CHEMTREC - Chile)
1 703 / 741-5970 (CHEMTREC - International)

Medical Emergency Number : Chile: CITUC: +56 2 2635 3800 (24 hours)

SECTION 2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

Label elements

Hazard pictograms :



Signal Word : WARNING

Hazard Statements : H410 Very toxic to aquatic life with long lasting effects.

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Precautionary Statements : **Prevention:**
P273 Avoid release to the environment.
Response:
P391 Collect spillage.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards
None known.

SECTION 3. COMPOSITION AND INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Systematic chemical name	Common Name	CAS-No.	Concentration or range (% w/w)	Classification
zinc oxide	Zinc oxide	1314-13-2	$\geq 30 - < 50$	Short-term (acute) aquatic hazard, Category 1 Long-term (chronic) aquatic hazard, Category 1
ethanediol	Ethylene glycol	107-21-1	$\geq 1 - < 5$	Acute toxicity (Oral), Category 4 Specific target organ toxicity - repeated exposure (Oral) (Kidney), Category 2
1,2-benzisothiazol-3(2H)-one	1,2-Benzisothiazol-3(2H)-one	2634-33-5	$\geq 0,0025 - < 0,025$	Acute toxicity (Oral), Category 4 Skin corrosion/irritation, Category 2 Serious eye damage/eye irritation, Category 1 Skin sensitization, Category 1 Short-term (acute) aquatic hazard, Category 1 Long-term (chronic) aquatic hazard, Category 2

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SECTION 4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- Inhalation : Move to fresh air.
If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.
- Skin contact : Take off all contaminated clothing immediately.
Wash contaminated clothing before re-use.
Wash off immediately with plenty of water for at least 15 minutes.
Get medical attention immediately if irritation develops and persists.
- Eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- Ingestion : Do not induce vomiting without medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
- Most important symptoms and effects, both acute and delayed : None known.
- Protection of first-aiders : First Aid responders should pay attention to self-protection and use the recommended protective clothing
Avoid inhalation, ingestion and contact with skin and eyes.
If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

- Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.
- Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.
- Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Carbon oxides
metal fumes

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| Related specific hazards | : | Do not allow run-off from fire fighting to enter drains or water courses. |
| Specific extinguishing methods | : | Remove undamaged containers from fire area if it is safe to do so.
Use a water spray to cool fully closed containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| Recommendations for fire-fighters | : | Firefighters should wear protective clothing and self-contained breathing apparatus. |

SECTION 6. ACCIDENTAL RELEASE MEASURES

- | | | |
|---|---|--|
| Personal precautions, protective equipment and emergency procedures | : | Evacuate personnel to safe areas.
Use personal protective equipment.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.
For disposal considerations see section 13. |
| Environmental precautions | : | Prevent further leakage or spillage if safe to do so.
Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods and material for containment and cleaning up | : | Never return spills in original containers for re-use.
Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labeled containers.
Keep in suitable, closed containers for disposal. |

SECTION 7. HANDLING AND STORAGE

Handling

- | | | |
|-------------------------------|---|---|
| Precautions for safe handling | : | Do not breathe vapors/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations. |
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Operational and technical measures : Normal measures for preventive fire protection.

Contact prevention : Avoid contact with skin, eyes and clothing.
Do not inhale aerosol.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

Conditions for safe storage, including any incompatibilities

Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.

Recommended storage temperature : > 5 °C

Further information on storage stability : No decomposition if stored and applied as directed.
Do not freeze.

Specific end use(s)

Specific use(s) : Crop nutrition

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible maximum concentration	Basis
zinc oxide	1314-13-2	LPP (Fumes)	4,4 mg/m ³	CL OEL
		LPT (Fumes)	10 mg/m ³	CL OEL
		TWA (Respirable particulate matter)	2 mg/m ³	ACGIH
		STEL (Respirable particulate matter)	10 mg/m ³	ACGIH
ethanediol	107-21-1	LPA (aerosol)	40 ppm 100 mg/m ³	CL OEL
		Further information: Substances classified as 'A.4' are under consideration but have not yet valid information to be classified as carcinogenic to		

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		humans or laboratory animals, so the exposure of workers to them should be maintained at the lowest possible level.		
		TWA (Vapor)	25 ppm	ACGIH
		STEL (Vapor)	50 ppm	ACGIH
		STEL (Inhalable fraction, Aerosol only)	10 mg/m ³	ACGIH

Personal protective equipment

- Eye/face protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Face-shield
- Skin protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.
- Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.
- Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.
- Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.
- Protective measures : Plan first aid action before beginning work with this product.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

- Physical state : liquid
- Form : suspension
- Color : white
- Odor : Faint odour
- Odor Threshold : No data available
- pH : 8,0 - 12,0
Concentration: 100 %
- Melting point/ range : No data available

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Boiling point/boiling range : No data available

Flash point : No data available

Evaporation rate : No data available

Upper explosion limit / Upper flammability limit : No data available

Lower explosion limit / Lower flammability limit : No data available

Vapor pressure : No data available

Vapor density : No data available

Relative density : No data available

Density : No data available

Solubility(ies)
Water solubility : No data available

Partition coefficient: n-octanol/water : No data available

Autoignition temperature : No data available

Decomposition temperature : No data available

Viscosity
Viscosity, dynamic : No data available

Viscosity, kinematic : No data available

Explosive properties : Not explosive

Oxidizing properties : Non-oxidizing

Other information

Bulk density : No data available

Molecular weight : Not applicable

Particle size : No data available

Self-ignition : No data available

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SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed.
Conditions to avoid	:	Avoid extreme temperatures. Avoid formation of aerosol. Protect from frost, heat and sunlight.
Incompatible materials	:	Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	:	toxic fumes

SECTION 11. TOXICOLOGICAL INFORMATION

Acute toxicity

Based on available data, the classification criteria are not met.

Product:

Acute oral toxicity	:	Acute toxicity estimate: > 2.000 mg/kg Method: Calculation method
Acute inhalation toxicity	:	Acute toxicity estimate: > 20 mg/l Exposure time: 4 h Test atmosphere: vapor Method: Calculation method

Components:

zinc oxide:

Acute oral toxicity	:	LD50 (Rat, male and female): > 2.000 mg/kg Method: OECD Test Guideline 423 LD50 (Mouse, male and female): > 2.000 mg/kg Method: OECD Test Guideline 401 Target Organs: Liver, Heart, spleen, Stomach, Pancreas Symptoms: Damage Remarks: mortality
Acute inhalation toxicity	:	LC50 (Rat, male and female): > 1,79 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: EPA OPP 81 - 3 Remarks: no mortality

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Acute dermal toxicity : LD50 Dermal (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402

ethanediol:

Acute inhalation toxicity : LC50 (Rat, male and female): > 2,5 mg/l
Exposure time: 6 h
Test atmosphere: dust/mist
Remarks: no mortality

Acute dermal toxicity : LD50 (Mouse, male and female): > 3.500 mg/kg

1,2-benzisothiazol-3(2H)-one:

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Remarks : No data available

Components:

zinc oxide:

Species : reconstructed human epidermis (RhE)
Method : OECD Test Guideline 431
Result : No skin irritation

ethanediol:

Species : Rabbit
Result : No skin irritation

1,2-benzisothiazol-3(2H)-one:

Species : Rabbit
Exposure time : 72 h
Method : OECD Test Guideline 404
Result : No skin irritation

Serious eye damage or eye irritation

Based on available data, the classification criteria are not met.

Product:

Remarks : No data available

Components:

zinc oxide:

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Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	No eye irritation

ethanediol:

Species	:	Rabbit
Result	:	No eye irritation

1,2-benzisothiazol-3(2H)-one:

Species	:	Bovine cornea
Method	:	OECD Test Guideline 437
Result	:	No eye irritation

Species	:	Rabbit
Method	:	EPA OPP 81-4
Result	:	Irreversible effects on the eye

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Remarks	:	No data available
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Components:

zinc oxide:

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Does not cause skin sensitization.

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Substance is not considered to be potential skin sensitizer.

ethanediol:

Test Type	:	Maximization Test
Species	:	Guinea pig
Result	:	Does not cause skin sensitization.

1,2-benzisothiazol-3(2H)-one:

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406

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Result : May cause sensitization by skin contact.

Species : Guinea pig

Method : FIFRA 81.06

Result : May cause sensitization by skin contact.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

zinc oxide:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: equivocal

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster fibroblasts
Method: OECD Test Guideline 473
Result: negative

Test Type: Chromosome aberration test in vitro
Test system: Human lymphocytes
Result: positive

Test Type: Micronucleus test
Test system: Human epithelioid cells
Method: OECD Test Guideline 487
Result: negative

Test Type: Micronucleus test
Test system: Human lymphocytes
Result: positive

Genotoxicity in vivo : Test Type: In vivo micronucleus test
Species: Mouse (male)
Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative

ethanediol:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OPPTS 870.5100
Result: negative

Genotoxicity in vivo : Test Type: dominant lethal test
Species: Rat
Application Route: Oral

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Result: negative

1,2-benzisothiazol-3(2H)-one:

Genotoxicity in vitro : Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: positive

Genotoxicity in vivo : Test Type: unscheduled DNA synthesis assay
Species: Rat (male)
Cell type: Liver cells
Application Route: Ingestion
Exposure time: 4 h
Method: OECD Test Guideline 486
Result: negative

Test Type: Micronucleus test
Species: Mouse
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

zinc oxide:

Species : Mouse, male and female
Application Route : Oral
Exposure time : 1 year
Dose : 4400, 22000 mg/l
NOAEL : > 22.000 mg/l
Result : negative
Remarks : Based on data from similar materials

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

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ethanediol:

Species	: Mouse
Application Route	: Oral
Exposure time	: 24 month(s)
Result	: negative

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

zinc oxide:

Effects on fertility	: Test Type: Two-generation study Species: Rat, male and female Application Route: Oral Dose: 7.5, 15, 30mg/kg bw/day Frequency of Treatment: 7 days/week General Toxicity Parent: LOAEL: 7,5 mg/kg body weight General Toxicity F1: LOAEL: 30 mg/kg body weight Method: OECD Test Guideline 416 Result: negative Remarks: Based on data from similar materials Test Type: one-generation reproductive toxicity Species: Rat, male Application Route: Oral Dose: 4,000 milligram per liter Frequency of Treatment: 32 daily General Toxicity Parent: LOAEL: 4.000 mg/l General Toxicity F1: LOAEL: 4.000 mg/l Symptoms: Reduced fertility Target Organs: male reproductive organs Result: positive Remarks: Based on data from similar materials
Effects on fetal development	: Species: Rat Application Route: inhalation (dust/mist/fume) Dose: .0003, 0.002, 0.008 milligram per liter Duration of Single Treatment: 14 d General Toxicity Maternal: LOAEC: 0,008 mg/L Developmental Toxicity: NOAEC: 0,008 mg/L Embryo-fetal toxicity.: NOAEC Mating/Fertility: 0,008 mg/L Method: OECD Test Guideline 414 Result: negative

1,2-benzisothiazol-3(2H)-one:

Effects on fertility	: Species: Rat, male Application Route: Ingestion General Toxicity Parent: NOAEL: 18,5 mg/kg body weight General Toxicity F1: NOAEL: 48 mg/kg body weight Fertility: NOAEL: 112 mg/kg bw/day Symptoms: No effects on reproduction parameters. Method: OPPTS 870.3800
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Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

Specific particular organ toxicity - single exposure

Based on available data, the classification criteria are not met.

Specific particular organ toxicity - repeated exposure

Based on available data, the classification criteria are not met.

Components:

ethanediol:

Routes of exposure : Oral
Target Organs : Kidney
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

1,2-benzisothiazol-3(2H)-one:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

zinc oxide:

Species : Rat, male and female
NOAEL : 31,52 mg/kg
LOAEL : 127,52 mg/kg
Application Route : Oral
Exposure time : 13 weeks
Dose : 0, 31.52, 127.52 mg/kg
Method : OECD Test Guideline 408
Target Organs : Pancreas
Symptoms : Necrosis
Remarks : Based on data from similar materials

Species : Mouse, male and female
NOEL : 3000 ppm
Application Route : Oral
Exposure time : 13 weeks
Dose : 0, 300, 3000, 30000 ppm
Method : OECD Test Guideline 408
Remarks : Based on data from similar materials

Species : Rat, male
LOAEL : 0,0045 mg/l
Application Route : inhalation (dust/mist/fume)
Exposure time : 3 months
Dose : 0.0003, 0.0015, 0.004mg/l
Method : OECD Test Guideline 413

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Target Organs	: Lungs
Remarks	: mortality
Species	: Rat, male and female
LOAEL	: 75 mg/kg bw/day
Application Route	: Dermal
Exposure time	: 28d
Dose	: 0, 75, 180, 360 mg/kg bw/day
Method	: OECD Test Guideline 410

ethanediol:

Species	: Rat
NOAEL	: 150 mg/kg
Application Route	: Oral
Exposure time	: 12 Months
Species	: Dog
NOAEL	: > 2.200 - < 4.400 mg/kg
Application Route	: Dermal
Exposure time	: 4 Weeks
Method	: OECD Test Guideline 410

1,2-benzisothiazol-3(2H)-one:

Species	: Rat, male and female
NOAEL	: 15 mg/kg
Application Route	: Ingestion
Exposure time	: 28 d
Method	: OECD Test Guideline 407
Symptoms	: Irritation
Species	: Rat, male and female
NOAEL	: 69 mg/kg
Application Route	: Ingestion
Exposure time	: 90 d
Symptoms	: Irritation, Reduced body weight

Inhalation hazard

Based on available data, the classification criteria are not met.

Experience with human exposure

Components:

zinc oxide:

Inhalation	: Symptoms: Fatigue, Sweating, bitter taste, chills, dry mouth, flu-like symptoms
Ingestion	: Symptoms: Gastrointestinal discomfort

Further information

Product:

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Remarks : No data available

SECTION 12. ECOLOGICAL INFORMATION

Toxicity

Components:

zinc oxide:

Toxicity to fish	: LC50 (Danio rerio (zebra fish)): 1,55 mg/l Exposure time: 96 h Test Type: static test
Toxicity to daphnia and other aquatic invertebrates	: LC50 (Daphnia magna (Water flea)): 0,76 mg/l Exposure time: 48 h Method: OECD Test Guideline 202 LC50: 0,37 mg/l Exposure time: 96 h Test Type: static test EC50: 0,14 mg/l Exposure time: 24 h Test Type: static test EC50: 0,072 mg/l Exposure time: 96 h Test Type: static test
Toxicity to algae/aquatic plants	: IC50 (Pseudokirchneriella subcapitata (algae)): 0,044 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 NOEC (Pseudokirchneriella subcapitata (algae)): 0,024 mg/l Exposure time: 3 d Method: OECD Test Guideline 201 IC50 (Skeletonema costatum (marine diatom)): 1,23 mg/l Exposure time: 96 h Method: OECD Test Guideline 201 IC50: 3,28 mg/l Exposure time: 96 h Method: OECD Test Guideline 201 NOEC (Dunaliella tertiolecta (marine algae)): 0,01 mg/l Exposure time: 4 d Test Type: static test EC50 (Dunaliella tertiolecta (marine algae)): 0,65 mg/l Exposure time: 4 d Test Type: static test

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(*Chlorella vulgaris* (Fresh water algae)): 1,16 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

EC50 (*Anabaena flos-aquae* (cyanobacterium)): 0,3 mg/l
Exposure time: 96 h
Test Type: static test

EC50: 0,69 mg/l
Exposure time: 3 d
Test Type: static test

EC50 (*Phaeodactylum tricornutum*): 1,12 mg/l
Exposure time: 24 h
Test Type: static test

M-Factor (Acute aquatic toxicity) : 1

Toxicity to microorganisms : EC50 (activated sludge): > 1.000 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209

EC50 (*Tetrahymena pyriformis*): 7,1 mg/l
Exposure time: 24 h
Test Type: Growth inhibition

Toxicity to fish (Chronic toxicity) : NOEC: 0,440 mg/l
Exposure time: 72 d
Species: *Oncorhynchus mykiss* (rainbow trout)
Test Type: flow-through test
Remarks: Based on data from similar materials

NOEC: 0,026 mg/l
Exposure time: 30 d
Species: *Jordanella floridae* (flagfish)
Method: OECD Test Guideline 210
Remarks: Based on data from similar materials

NOEC: 0,530 mg/l
Exposure time: 1.095 d
Species: *Salvelinus fontinalis* (Brook trout)
Test Type: flow-through test
Remarks: Based on data from similar materials

NOEC: 0,056 mg/l
Exposure time: 116 d
Species: *Salmo trutta* (brown trout)
Method: OECD Test Guideline 210
Remarks: Based on data from similar materials

NOEC: 0,025 mg/l
Exposure time: 27 d
Species: Fish

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Test Type: semi-static test
Remarks: Based on data from similar materials

NOEC: 0,078 mg/l
Exposure time: 248 d
Species: Pimephales promelas (fathead minnow)
Test Type: flow-through test
Remarks: Based on data from similar materials

NOEC: 0,050 mg/l
Exposure time: 155 d
Species: Fish
Test Type: flow-through test
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : LOEC: 0,125 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Method: OECD Test Guideline 211

M-Factor (Chronic aquatic toxicity) : 10

Toxicity to soil dwelling organisms : NOEC: 750 mg/kg
Exposure time: 21 d
Species: Eisenia fetida (earthworms)

ethanediol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): > 72.860 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : IC50 (Pseudokirchneriella subcapitata (green algae)): 10.940 mg/l
Exposure time: 96 h

Toxicity to microorganisms : (activated sludge): > 1.995 mg/l
Exposure time: 30 min
Method: ISO 8192

Toxicity to fish (Chronic toxicity) : 1.500 mg/l
Exposure time: 28 d
Species: Menidia peninsulae (tidewater silverside)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : 33.911 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)

1,2-benzisothiazol-3(2H)-one:

Toxicity to fish : LC50 (Cyprinodon variegatus (sheepshead minnow)): 16,7

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mg/l
Exposure time: 96 h
Test Type: static test

LC50 (Oncorhynchus mykiss (rainbow trout)): 2,15 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 2,9 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 0,070 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Pseudokirchneriella subcapitata (green algae)): 0,04 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

M-Factor (Acute aquatic toxicity) : 10

Toxicity to microorganisms : EC50 (activated sludge): 24 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

EC50 (activated sludge): 12,8 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

Persistence and degradability

Components:

zinc oxide:

Biodegradability : Remarks: The methods for determining the biological degradability are not applicable to inorganic substances.

ethanediol:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 90 - 100 %
Exposure time: 10 d
Method: OECD Test Guideline 301A

1,2-benzisothiazol-3(2H)-one:

Biodegradability : Result: rapidly biodegradable

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Method: OECD Test Guideline 301C

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

zinc oxide:

Bioaccumulation : Species: Oncorhynchus mykiss (rainbow trout)
Exposure time: 14 d
Bioconcentration factor (BCF): 2.060
Remarks: Bioaccumulation is unlikely.

ethanediol:

Partition coefficient: n-octanol/water : log Pow: -1,36

1,2-benzisothiazol-3(2H)-one:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Exposure time: 56 d
Bioconcentration factor (BCF): 6,62
Method: OECD Test Guideline 305
Remarks: Substance is not persistent, bioaccumulative, and toxic (PBT).

Partition coefficient: n-octanol/water : log Pow: 0,7 (20 °C)
pH: 7

log Pow: 0,99 (20 °C)
pH: 5

Mobility in soil

Components:

1,2-benzisothiazol-3(2H)-one:

Distribution among environmental compartments : Koc: 9,33 ml/g, log Koc: 0,97
Method: OECD Test Guideline 121
Remarks: Highly mobile in soils

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

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SECTION 13. DISPOSAL CONSIDERATIONS

Waste treatment methods

- Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.
- Contaminated packaging, and contaminated material : It is prohibited to reuse, bury, burn, or sell containers. Rinsable containers: Triple rinse containers of less than 20 liters and pressure rinse containers of 20 liters or more. Triple rinsing: Add water up to $\frac{1}{4}$ of the container's capacity, close and shake for 30 seconds. Pour the rinse water into the mixing tank, considering this volume of water within the recommended volume for mixing preparation. Perform this procedure three times. Pressure rinsing: Activate the pressure rinsing device for 30 seconds, considering the volume of water used as part of the recommended volume for mixing preparation. In both procedures, punctured the container on its base without damaging the label. In all cases, take the empty containers to collection points indicated by the local empty containers program.

SECTION 14. TRANSPORT INFORMATION

International Regulations

UNRTDG

- UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Zinc oxide)
Class : 9
Packing group : III
Labels : 9
Environmentally hazardous : yes

IATA-DGR

- UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(Zinc oxide)
Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code

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UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Zinc oxide)
Class	:	9
Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F
Marine pollutant	:	yes

Transport in bulk according to IMO instruments

Not applicable for product as supplied.

Domestic regulation

NCh382

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Zinc oxide)
Class	:	9
Packing group	:	III
Labels	:	9
Environmentally hazardous	:	yes

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

National Regulations

Chile. Decree 190. Carcinogenic Substances, Hazardous Waste Management. : Not applicable

Decree 1358 - Establishment of rules governing the control measures of precursors and essential chemicals. : sodium hydroxide

Resolution 408/16 Exempt, Approving List of Health Hazardous Substances : Included in list of Article 3, item a), Classification according to NCh382

Other regulations

Decree 43/2015, Approving Regulation on Storage of Hazardous Substances
NCh 2245:2021 Safety data sheet for chemical products - Content and order of sections
NCh 2190:2019 Land transport of dangerous goods - Hazard identification marks
NCh 382:2021 Dangerous Goods – Classification
Decree 57 of 2019, Regulation on Classification, Labeling, and Notification of Hazardous Chemicals and Mixtures
D.S. 148/03 Sanitary Regulation on hazardous wastes handling
D.S. 298/94 Regulation on transport of hazardous cargo on streets and roads
D.S. 594/99 Regulation on sanitary and environmental basic conditions at work places

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Exempt Resolution 15 of 2023 approving the List of Hazardous Substances Subject to Import Process

The ingredients of this product are reported in the following inventories:

TCSI	:	Not in compliance with the inventory
TSCA	:	Product contains substance(s) not listed on TSCA inventory.
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains the following components that are not on the Canadian DSL nor NDSL. Sodium Polyacrylate Homopolymer
ENCS	:	Not in compliance with the inventory
ISHL	:	Not in compliance with the inventory
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

The receiver should verify the possible existence of legal regulations applicable to chemical.

SECTION 16. OTHER INFORMATION

Revision Date : 18.12.2025

Date format : dd.mm.yyyy

Full text of H-Statements

Abbreviations and acronyms

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Serious eye damage/eye irritation	:	Serious eye damage/eye irritation
Skin corrosion/irritation	:	Skin corrosion/irritation
Skin Sens.	:	Skin sensitization
STOT RE	:	Specific target organ toxicity - repeated exposure
ACGIH	:	USA. ACGIH Threshold Limit Values (TLV)
CL OEL	:	Chile. Regulation on basic sanitary and environmental conditions in the workplace

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ACGIH / TWA	:	8-hour, time-weighted average
ACGIH / STEL	:	Short-term exposure limit
CL OEL / LPP	:	Time Weighted Limit Value
CL OEL / LPT	:	Short Term Limit Value
CL OEL / LPA	:	Ceiling value

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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